Topics to Review for DS6021 Midterm

While everything we have covered so far this semester is fair game, if you focus on the topics listed below, you should be in good shape.

1. Bias-variance tradeoff and model complexity
    a. What is the bias variance tradeoff?
    b. How do bias and variance change across different types of models? (e.g. Linear Regression, Polynomial Regression, Splines)
    c. How do training and testing error change when you increase variance?
2. Linear Regression
    a. Basics of OLS and being able to identify a simple linear regression model
    b. Reading a coefficient table (similar to what we have seen in statsmodels)
    c. What is VIF and what does it tell us?
    d. What does R^2 tell us about model fit?
3. Cross-validation
    a. Why is cross-validation important?
    b. Why is cross-validation better than a single train/test split?
    c. How does cross-validation impact bias in estimated predicted error?
4. Categorical variables and interaction terms
    a. How categorical variables are encoded for either linear or logistic regression.
    b. How to express an interaction term as part of a linear regression
    c. How an interaction term changes the interpretation of the beta coefficients.
    d. The first two questions on Quiz #2 will be helpful in preparing.
5. Logistic regression
    a. Basics of logistic regression: The logit function, odds ratio, log odds.
    b. Log loss and how it penalizes "confidently wrong predictions"
    c. Why accuracy is a bad metric.
    d. Other metrics that are used to assess logistic regression models: recall, precision, F1 score, AUC.
6. Regularization
    a. Goals of regularization
    b. How does regularization treat correlated features?
    c. Which types of regularization can be used for feature selection and why?
7. Sklearn pipelines
    a. Different stages of a preprocessing pipeline
    b. What different types of encoders do
    c. Why we do preprocessing on training data only prior to fitting?

Things that will not be on the Midterm:

- A/B Testing
- Sample Comparisons
- Bootstrapping
- Forward and Backward Feature Selection, including Recursive Feature Elimination

Format of the Midterm:

- 8 total questions, each with multiple parts
- 50 minutes to complete the midterm
- You should be able to complete each question in about 6 minutes